

SOCIAL NETWORK ANALYSIS

Introduction

This section analyses the Facebook social network dataset from the SNAP Large Network Dataset Collection. The data is provided as an edge list, with each row representing a friendship connection between anonymised users. Using NetworkX, the graph is built and examined to understand its overall structure. The aim of this analysis is to describe the network structure and identify key patterns in how users are connected. Network measures (nodes, edges, density, and degree distribution) are analysed, alongside edge betweenness centrality and community detection (Girvan–Newman and Label Propagation) to highlight important links and friendship groups within the network.

Analysis and Results

The Facebook edge list was used to build an undirected graph in NetworkX. The network contains 4,039 nodes and 88,234 edges, with a low density of 0.0108, showing that the network is sparse. However, the average degree (43.69) indicates that users still have many connections on average, meaning friendships are concentrated within smaller groups rather than spread across the whole network.

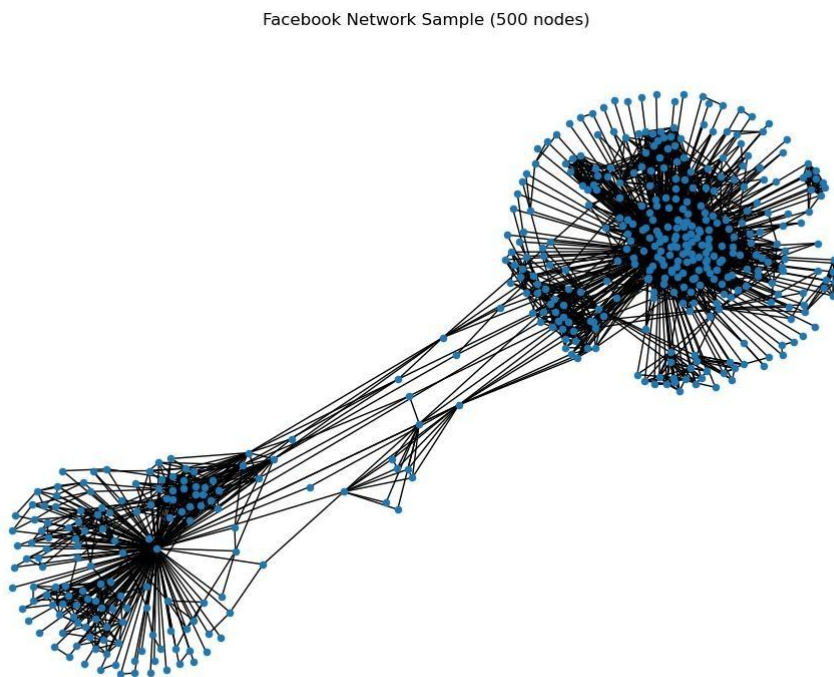


Figure 1: Facebook Network Visualisation with 500 Node Sample

Figure 1 shows the sampled network structure. The plot shows visible clusters of connected users, suggesting the presence of friendship circles. A degree distribution plot also shows that most users have few connections, while a small number have many connections, indicating hub-like behaviour, which is common in social networks.

According to Kim et al. (2025), this metric measures the frequency with which an edge is on the shortest path between nodes. As a result, edges with high values can serve as links between various network components.

To identify important connections, edge betweenness centrality was calculated. This measure captures how repeatedly an edge lies on the shortest path between nodes, meaning edges with high values can serve as links between various parts of the network (Kim et al., 2025). Figure 2 shows that majority edges have centrality values close to zero, with only few having significantly higher values. These high-centrality edges likely act as key connectors between different groups.

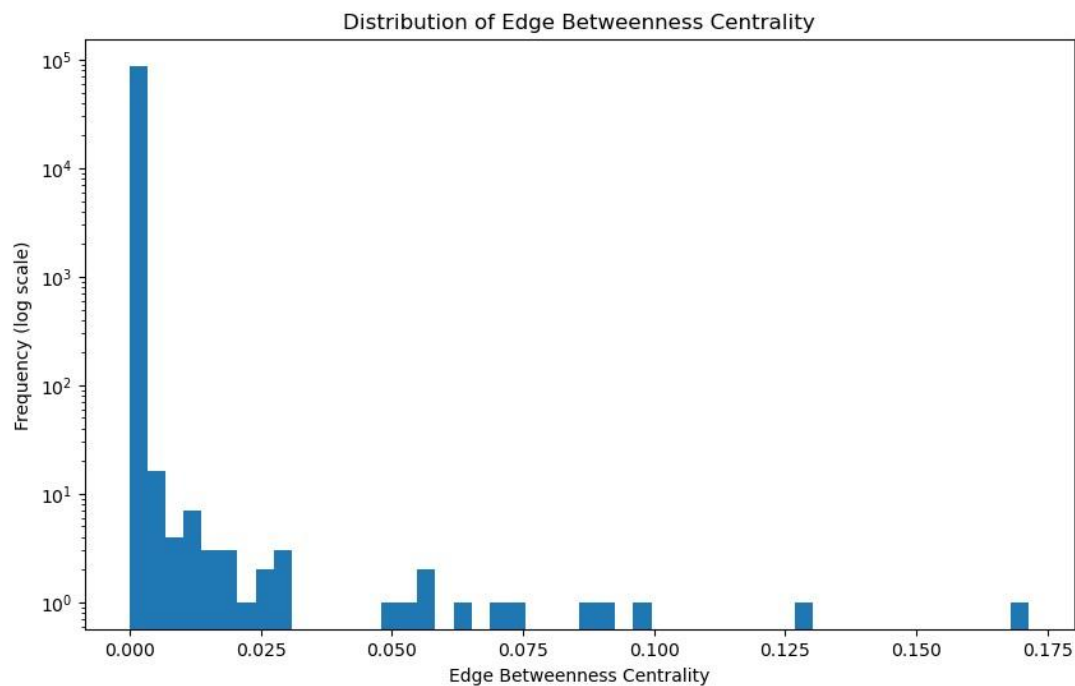


Figure 2: Histogram Distribution of the Edge Centrality

Community detection was used to examine clustering in the network using two algorithms, compared based on number and sizes of communities. Girvan–Newman detects communities by iteratively removing edges with high betweenness centrality (Kumar and Singh, 2024). Due to its high computational cost, it was applied to a 500-node sample (3,513 edges). It produced 2 communities at the first split (sizes 352 and 148) and 3 communities at the second split (343, 148, 9), as shown in Figure 3.

Girvan-Newman Communities (500 nodes)

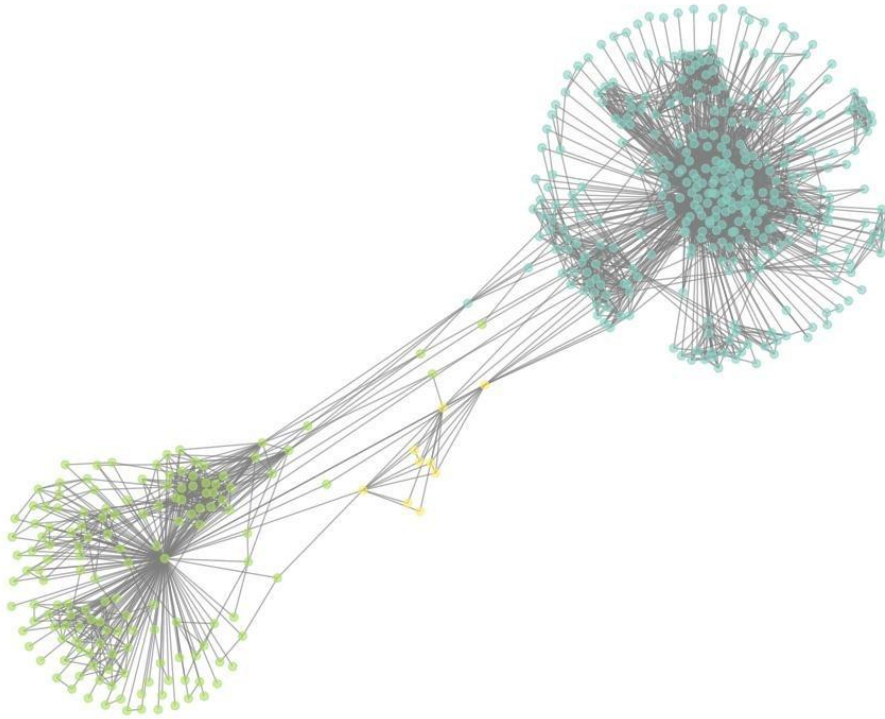


Figure 3: Girvan-Newman Visualisation of 3 Communities with 500 Node Sample

Label Propagation was applied to the full network because it is efficient for large graphs and forms communities by spreading labels through connected nodes (Traag and Šubelj, 2022). It detected 44 communities, ranging from large groups (e.g., 1,030, 753, 547, 469 nodes) to many small communities with fewer than 10 nodes (Figure 4). This suggests the Facebook network contains many groups of different sizes, rather than only a few large clusters.

Label Propagation Communities (500 nodes)

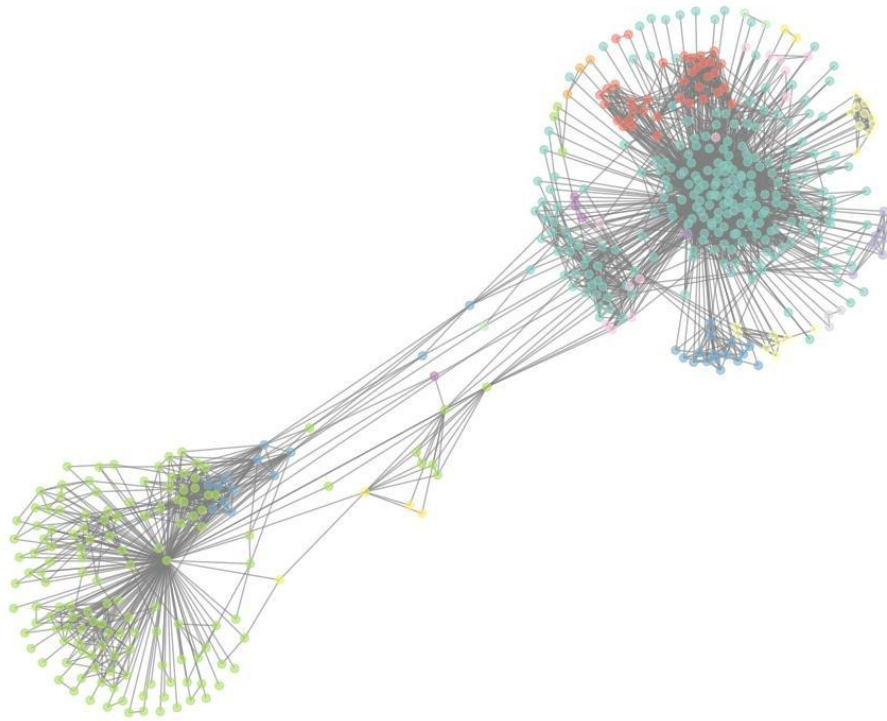


Figure 4: Label Propagation Visualisation of Communities with 500 Node Sample

Discussion

The results show that the Facebook friendship network is sparse overall, with a low density, meaning most users are not directly connected to most other users. However, the high average degree suggests that users still maintain many friendships, but mainly within smaller groups rather than across the entire network. This supports the idea that Facebook connections tend to form clusters of friends instead of one fully connected community.

The degree distribution revealed that majority of users have few connections, while a small percentage have many. These highly connected users likely act as hubs, helping to link people within and across groups. In a real social setting, these hubs may represent individuals with wider social circles who can influence how information spreads through the network.

Edge betweenness centrality revealed that most edges have very low values, meaning most friendships exist within local groups and do not strongly affect the network's overall connectivity. However, a small number of edges had high centrality, suggesting they act as bridge connections between different friendship circles. Removing or weakening these links could reduce connectivity between communities and make the network more disconnected.

Community detection confirmed that the network contains clear clustering. Girvan–Newman produced a small number of broad communities in the sampled network, which shows that it tends to detect large, high-level divisions first. As the algorithm continues splitting, smaller groups begin to appear. In contrast, Label Propagation detected many communities across the full network, capturing a more detailed view of how friendship circles form at different sizes. Overall, the results suggest the Facebook network is organised into multiple communities with a few important users and connections helping to maintain links between them.

Limitations: Girvan–Newman was run on a 500-node sample due to its high computational cost, so its community results may not fully represent the full 4,039-node network. In addition, Label Propagation can produce slightly different communities when re-run because it may not be fully deterministic.

Conclusion

This analysis constructed a Facebook friendship network with 4,039 nodes and 88,234 edges. Although the network is sparse due to its low density, the high average degree confirmed that users still have many connections within smaller groups. Edge betweenness centrality highlighted that most connections are local, while a few act as important bridges between groups. Community detection confirmed clear clustering, with Girvan–Newman identifying broad community splits and Label Propagation revealing many friendship circles of different sizes. Overall, the Facebook network is organised into clustered communities connected through a small number of key users and links.

References

- Traag, V.A. and Šubelj, L., 2023. Large network community detection by fast label propagation. *Scientific Reports*, 13(1), p.2701.
- Kim, J., Jeong, S., Kim, J. and Lim, S., 2025. Bridges in social networks: current status and challenges. *PeerJ Computer Science*, 11, p.e3122.
- Kumar, P. and Singh, D., 2024. The evaluation of community detection techniques on real-world networks. *Social Network Analysis and Mining*, 14(1), p.162.